

Few-shot industrial defect detection using vision-language models (e.g., YOLO-World) to overcome cold-start data scarcity in automated surface inspection for manufacturing.

Manufacturing × Automated quality inspection × Machine learning · OS023 v3 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
45 is the world moving	74 can we play, can we win	MEDIUM 0 named in the evidence	€9.3m observed evidence · per year	LATER research and standards activi...	L1 18 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

The opportunity

This opportunity is about deploying few-shot industrial defect detection using vision-language models like YOLO-World to overcome the cold-start problem in automated surface inspection for manufacturing. It targets manufacturers facing data scarcity for new defect types, enabling faster deployment of quality inspection systems. The need is driven by recent research demonstrating the viability of adapting VLMs for few-shot defect detection, addressing a long-standing challenge in the industry.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€12.3m €2.2m – €66.1m	€9.3m €1.7m – €50.2m	€449k €81k – €2.4m	observed
Observed: tendered contract value, annualised	€20.9m €20.9m – €56.6m	€0	€0	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Manufacturing (DK)	2,685 enterprises	observed	Eurostat · sbs_sc_ovw	2024
Enterprises adopting Machine learning	7.3 % of enterprises (10+ employees)	observed	Eurostat · isoc_eb_ain2 · E_AI_TML	2025
Annual value of one engagement	€337k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Size-weighted buyer base	497 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	4.8 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders. 3 of 4 declared geographies are outside the European reference data (CN, IN, TW); the estimate covers the European footprint only.

Observed: tendered contract value, annualised

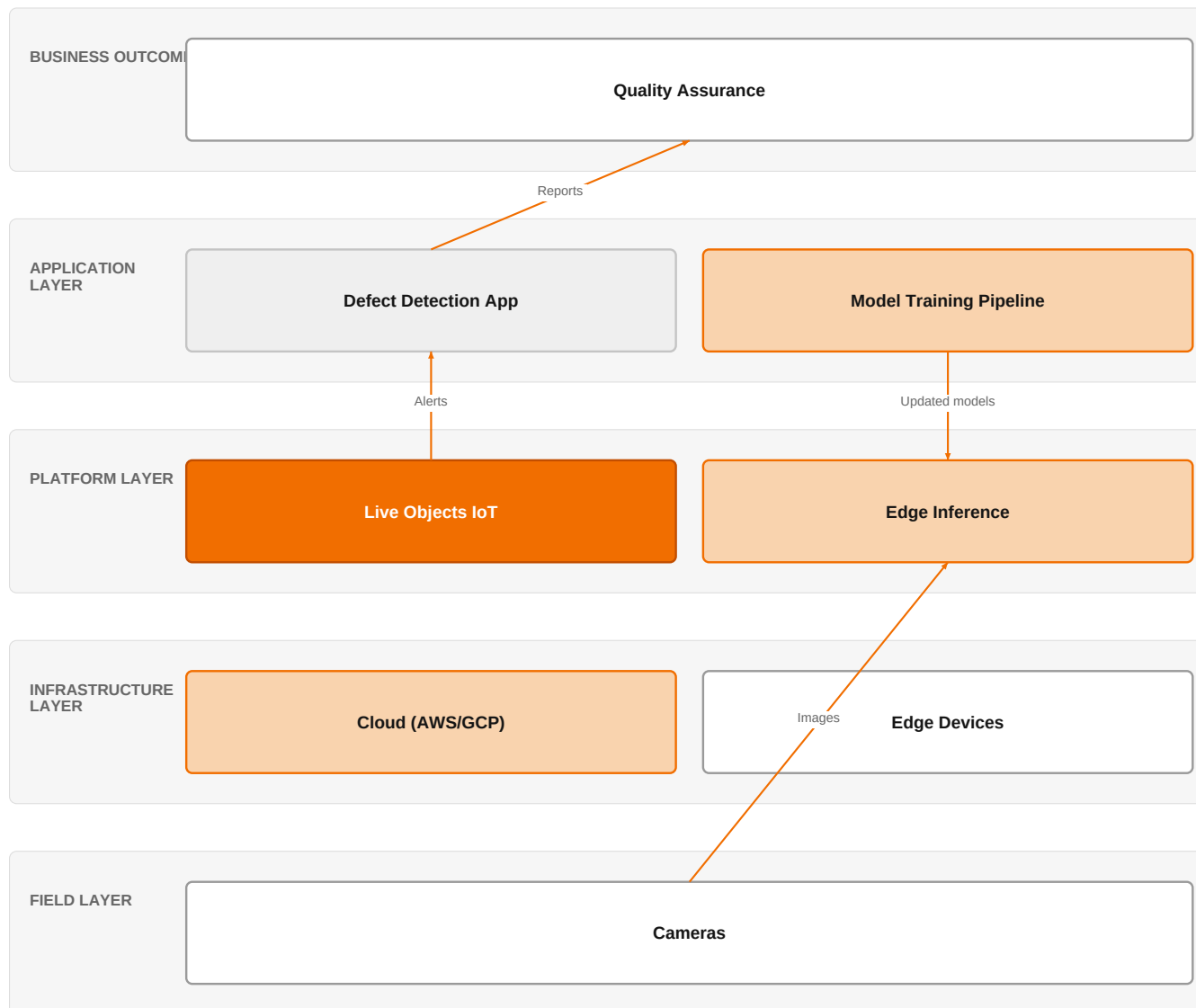
Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€20.9m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Assumed obtainable share of the serviceable market	4.8 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 14 matching notices, matched on vertical via the CPV crosswalk — \$4.5.2's warning applies: a crosswalk error lands directly

in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size. 3 of 4 declared geographies are outside the European reference data (CN, IN, TW); the estimate covers the European footprint only.

The solution, in one picture

Few-Shot Defect Detection Solution



This diagram shows how Orange integrates edge cameras, IoT platform, and cloud-based training to deliver few-shot defect detection.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

Recent research has shown that adapting vision-language models (VLMs) like YOLO-World can mitigate the cold-start problem in industrial defect detection, where limited annotated data for new anomalies hinders traditional detectors [SIG-8CA4FFD436BC]. This is part of a broader trend where deep learning and computer vision are being applied to various defect detection tasks, including welding [SIG-BCC58EC44BA3], casting [SIG-83B109035468], and additive manufacturing [SIG-A598AE09F54F], with a focus on real-time and edge deployment [SIG-5B350B1A3B15]. Additionally, the expansion of NIST's chemical fingerprint library supports manufacturers in identifying unknown compounds, indicating a regulatory and standards-driven push for quality assurance [SIG-7B84CE15615E].

Sources: SIG-8CA4FFD436BC openalex.org 2026-03-27; SIG-BCC58EC44BA3 openalex.org 2025-11-18; SIG-83B109035468 openalex.org 2025-12-03; SIG-A598AE09F54F openalex.org 2025-11-20; SIG-5B350B1A3B15 Journal of Organizatio 2026-05-21; SIG-7B84CE15615E nist.gov 2026-06-09

Who buys, and why now

The buyer is typically a manufacturing quality director or plant manager who is responsible for maintaining product quality and operational efficiency. The pain is the 'cold-start' problem: when a new product line or defect type emerges, there is insufficient annotated data to train a reliable detection model, leading to prolonged manual inspection and increased defect escape rates. The trigger is often a new product launch, a customer complaint about quality, or a regulatory audit that exposes gaps in inspection capabilities.

Why it is hot now — every claim bound to a dated source

- Adapting vision-language models enables few-shot industrial defect detection, mitigating the cold-start problem due to limited annotated data. [SIG-8CA4FFD436BC]
- Integrating deep learning with gated attention and squeeze-excitation fusion U-Net enables automated weld defect detection. [SIG-BCC58EC44BA3]
- Deep convolutional neural networks are used for casting defect detection in aluminium metal matrix composites. [SIG-83B109035468]
- Deep convolution neural networks can detect defects in aluminium metal matrix composite castings, a specific material and process. [SIG-83B109035468]
- Integrating deep learning into weld inspection addresses the inadequacy of manual methods for safety and precision. [SIG-BCC58EC44BA3]
- Aluminium metal matrix composite casting defect detection is performed using deep convolution neural networks. [SIG-83B109035468]
- Computer vision techniques are evaluated for quality inspection in casting manufacturing processes. [SIG-1EDDEA256F85]
- Deep convolutional neural networks are used for aluminium metal matrix composite casting defect detection. [SIG-83B109035468]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a solution leveraging its AI/Data/Cloud experts and IoT experts to deploy a few-shot defect detection system. The engagement would start with a proof-of-concept using the customer's existing camera infrastructure and data, applying vision-language models like YOLO-World to detect defects with minimal annotated samples. Orange would integrate the solution with its Live Objects IoT platform for edge data collection and management, and leverage partnerships with NVIDIA for edge hardware optimization and AWS or Google Cloud for scalable model training and deployment. The solution would be delivered on certified infrastructure (ISO 27001, TISAX) to meet security and compliance needs.

Why Orange can win

Orange can win by combining its expertise in AI/Data/Cloud and IoT with its certified infrastructure (ISO 27001, TISAX) to offer a secure, end-to-end solution. The partnership with NVIDIA enables optimized edge deployment, while the Live Objects platform provides a robust IoT backbone. However, the assets are thin: there is no specific reference for defect detection in manufacturing, and the core VLM technology is not proprietary. Orange's advantage lies in its ability to integrate and deploy at scale with security certifications, which is critical for industrial customers.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Live Objects	offer	L1	offer provides the technology in a matching domain	unconfirmed
AWS	partner	L2	partner provides the required technology	unconfirmed
Google Cloud	partner	L2	partner provides the required technology	unconfirmed
NiCE	partner	L2	partner provides the required technology	unconfirmed
NVIDIA	partner	L2	partner provides the required technology	unconfirmed
ISO 27001	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
TISAX	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
AI / Data / Cloud experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed

IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Colgate-Palmolive	reference	SUP	published reference in this vertical, different use case	unconfirmed
Mondelez International	reference	SUP	published reference in this vertical, different use case	unconfirmed
Saint-Gobain Glass	reference	SUP	published reference in this vertical, different use case	unconfirmed
Skytale	reference	SUP	published reference in this vertical, different use case	unconfirmed

6 further published reference(s) in this vertical are linked to this topic and are listed in the radar.

Competitive landscape

MEDIUM competitive intensity (score 7.1; 0 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

Competitor	Category	Position here	Basis	Evidence
AWS	Hyperscaler	sells Machine learning; competes in OX: Smart Industries — also an Orange partner	structural	—
Google Cloud	Hyperscaler	sells Machine learning — also an Orange partner	structural	—
Atos / Eviden	Global systems integrator	sells Machine learning	structural	—
NiCE	Customer-experience platform	sells Machine learning — also an Orange partner	structural	—
Databricks	Data and AI platform	sells Machine learning	structural	—
Dataiku	Data and AI platform	sells Machine learning	structural	—
Palantir	Data and AI platform	sells Machine learning; competes in OX: Smart Industries	structural	—
Accenture	Global systems integrator	active in Manufacturing; competes in OX: Smart Industries	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 When you introduce a new product or material, how long does it typically take to collect enough annotated defect images to train a reliable inspection model?
- 2 What is your current process for handling new defect types that were not present during initial model training?
- 3 Do you have existing camera and edge computing infrastructure on your production lines that could support a vision-based inspection system?
- 4 What are your data privacy and security requirements for deploying a cloud-based or edge-based ML solution?
- 5 Have you already evaluated any vision-language models or few-shot learning approaches for defect detection? If so, what were the outcomes?

Objections you will meet

They will say	You can answer
We already have a defect detection system in place; why would we switch?	That's a fair point. Many manufacturers have existing systems, but they often struggle with the cold-start problem when new defects appear. Our approach using vision-language models can adapt quickly with minimal data, reducing downtime and manual effort. We can complement your existing system or replace it, depending on your needs.

How do we know this will work on our specific defects and materials?	You're right to be cautious. The technology is proven in research, but real-world performance depends on your specific environment. That's why we propose a proof-of-concept using your data and cameras to demonstrate accuracy before any commitment. We'll also leverage our partnerships with NVIDIA and cloud providers to optimize for your edge devices.
We're concerned about data security and sovereignty.	Security is a top priority for us. We offer ISO 27001 and TISAX certified infrastructure, and we can deploy the solution on-premises or in a sovereign cloud. Our IoT platform, Live Objects, is designed with security in mind. We'll work with you to meet your compliance requirements.

Risks and unknowns

The main risk is that few-shot VLM-based detection may not yet be production-ready for all defect types, as research is still evolving. There is uncertainty about the accuracy and reliability of these models in real-world manufacturing environments with varying lighting, surfaces, and defect types. Additionally, the customer may already have investments in competitor platforms, making switching costly. The lack of a proven reference in this specific use case is a gap that needs to be addressed through a pilot.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief for a manufacturing client's automated quality inspection project, incorporating Orange's Live Objects offer, ISO 27001 and TISAX certifications, and partnerships with AWS, Google Cloud, and NVIDIA to address cold-start data scarcity.
Sales	In a conversation with a manufacturing quality manager, mention Orange's expertise in AI and IoT, and reference our work with Saint-Gobain Glass to illustrate our understanding of industrial inspection challenges.
Strategist / Innovator	Commission a deep-dive brief on few-shot industrial defect detection using vision-language models, leveraging Orange Group researchers and AI/Data/Cloud experts to assess feasibility and identify a pilot manufacturing use case.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-7B84CE15615E	2026-06-09	nist.gov	1	NIST Expands Its Library of 'Chemical Fingerprints' to Identify Unknown Substances
SIG-5B350B1A3B15	2026-05-21	Journal of Organizational...	1	Latency Trade-Offs of a Lightweight YOLOv11 Detection Algorithm on Edge Computing Nodes for Fac...
SIG-8CA4FFD436BC	2026-03-27	openalex.org	1	Adapting Vision–Language Models for Few-Shot Industrial Defect Detection
SIG-1D3047429B46	2025-12-12	openalex.org	1	A comprehensive survey for real-world industrial surface defect detection: Challenges, approach...
SIG-92E2E10DA219	2025-12-12	openalex.org	1	A real-time surface defect detection model based on adaptive feature information selection and ...
SIG-E0AA4F59988C	2025-12-10	openalex.org	1	Anomaly detection for industrial applications: challenges, solutions, and future directions
SIG-83B109035468	2025-12-03	openalex.org	1	Aluminium metal matrix composite casting defect detection using deep convolution neural networks
SIG-A598AE09F54F	2025-11-20	openalex.org	1	Real-time vision-based defect detection for large-scale on-site earthen additive manufacturing:...
SIG-BCC58EC44BA3	2025-11-18	openalex.org	1	Automated weld defect detection using gated attention and squeeze and excitation fusion U-Net
SIG-247B09FE185A	2025-11-07	openalex.org	1	Advancements in Small-Object Detection (2023–2025): Approaches, Datasets, Benchmarks, Applicati...
SIG-E9AF6D858220	2025-10-21	openalex.org	1	Review on Application of Machine Vision-Based Intelligent Algorithms in Gear Defect Detection
SIG-1EDDEA256F85	2025-10-16	openalex.org	1	Evaluation of computer vision techniques for quality inspection in casting manufacturing process
SIG-3C2F7CAC7510	2025-10-01	openalex.org	1	Federated multi scale vision transformer with adaptive client aggregation for industrial defect...
SIG-66E8F1ED640C	2025-09-18	openalex.org	1	Textile Defect Detection Using Artificial Intelligence and Computer Vision—A Preliminary Deep L...
SIG-D46EBCBB2C04	2025-09-16	openalex.org	1	Embedded vision transformers for industrial welding defect detection
SIG-81F8AA75428C	2025-09-09	openalex.org	1	Time-Series-Based Anomaly Detection in Industrial Control Systems Using Generative Adversarial ...
SIG-5DD84E684B5F	2025-08-21	openalex.org	1	Autonomous printing process optimisation and in-situ anomaly detection in fused deposition mode...
SIG-A06EDB8B4416	2026-04-22	arxiv.org	3	IoT-Enhanced CNN-Based Labelled Crack Detection for Additive Manufacturing Image Annotation in ...

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: competitive_landscape (no claim survived evidence binding); diagram (1 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:15+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:18+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-18T20:41:47+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.